Brutal Practice Paper B42

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. A freezer is at -8 C. It is cooled by 3 C each hour for 4 hours, then warmed by 5 C. What is the final temperature? [Integers]

- **(A)** 15 C
- **(B)** -3 C
- **(C)** -5 C
- **(D)** -20 C
- **(E)** -25 C
- **(F)** -15 C
- **(G)** -35 C
- **(H)** -11 C

2. Multiply 0.6 x 0.05. [A Peek Beyond the Point]

- **(A)** 0.05
- **(B)** 0.30
- **(C)** 0.035
- **(D)** 0.03
- **(E)** 0.003
- **(F)** 0.3
- **(G)** 3

3. The rusting of an iron gate is best described as which kind of change? [Physical & Chemical Changes]

- **(A)** Only a change of state
- **(B)** A chemical change that cannot be easily reversed
- **(C)** Only a change of shape
- **(D)** A chemical change that is easily reversed
- **(E)** A change that forms no new substance
- **(F)** A physical change that cannot be reversed
- **(G)** A physical change that is easily reversed

4. What is the key difference between weather and climate? [Weather, Climate & Adaptations]

- **(A)** Weather is the long-term average; climate is the daily change
- **(B)** They mean exactly the same thing
- **(C)** Weather is indoors; climate is outdoors
- **(D)** Weather is the day-to-day state of the atmosphere; climate is its average over many years
- **(E)** Climate is about one day; weather is about a century
- **(F)** Weather is measured only at night; climate only by day
- **(G)** Weather is only about rain; climate is only about heat

5. The product of two integers is -48 and one of them is 6. What is the SUM of the two integers? [Integers]

(A) 2
(B) -42
(C) -2
(D) 14
(E) 54
(F) 8
(G) -14

6. Evaluate 0.75 + 1.8 - 0.9. [A Peek Beyond the Point]

(A) 1.65
(B) 1.05
(C) 2.55
(D) 0.66
(E) 0.65
(F) 1.55
(G) 1.75

7. 12 g of magnesium burns and combines with oxygen to form 20 g of magnesium oxide. What mass of oxygen combined with the magnesium? [Physical & Chemical Changes]

(A) 4 g
(B) 8 g
(C) 12 g
(D) 16 g
(E) 20 g
(F) 32 g

8. On a day the maximum temperature is 34 C and the minimum is 22 C. What are the daily temperature range and the mean temperature? [Weather, Climate & Adaptations]

(A) Range 12 C, mean 28 C
(B) Range 12 C, mean 24 C
(C) Range 12 C, mean 56 C
(D) Range 56 C, mean 28 C
(E) Range 22 C, mean 34 C
(F) Range 12 C, mean 27 C

9. A bank account starts overdrawn at -250 rupees. Three deposits of 180 rupees each are made, then 95 rupees is withdrawn. What is the final balance? [Integers]

(A) 195 rupees
(B) 200 rupees
(C) 540 rupees
(D) -195 rupees
(E) 635 rupees
(F) 385 rupees
(G) 290 rupees

10. Apples cost 48 rupees per kg and grapes 120 rupees per kg. Find the cost of 1.5 kg apples and 0.25 kg grapes together. [A Peek Beyond the Point]

(A) 30 rupees
(B) 92 rupees
(C) 168 rupees
(D) 72 rupees
(E) 96 rupees
(F) 132 rupees
(G) 102 rupees

11. Why is the melting of wax considered a physical change? [Physical & Chemical Changes]

(A) It releases a new gas
(B) A completely new substance is formed
(C) The wax is still wax and can be solidified again
(D) Energy alone makes it a chemical change
(E) The wax can never be turned back to solid
(F) It changes colour into a new material

12. Which set of features helps a polar bear survive in its very cold habitat? [Weather, Climate & Adaptations]

(A) Thick fur, a layer of blubber, and white colour for camouflage
(B) A naked body like a desert animal
(C) A long bare tail to lose heat
(D) Thin skin and no body fat
(E) Webbed wings for flying
(F) Large ears to stay cool

13. Evaluate (-6) x (-4) + (-5) x 3. [Integers]

(A) 39
(B) 9
(C) -24
(D) 24
(E) 3
(F) -9

14. Divide 4.5 / 0.9. [A Peek Beyond the Point]

(A) 45
(B) 0.05
(C) 5
(D) 4.5
(E) 4.05
(F) 0.5

15. Which of the following is a chemical change? [Physical & Chemical Changes]

(A) Folding a sheet of paper
(B) Stacking sheets of paper
(C) Wetting a sheet of paper
(D) Tearing a sheet of paper
(E) Cutting paper into strips
(F) Crumpling paper into a ball
(G) Burning a sheet of paper

16. Which instrument is used to measure the amount of rainfall? [Weather, Climate & Adaptations]

(A) Wind vane
(B) Ruler on a wall
(C) Rain gauge
(D) Barometer
(E) Anemometer
(F) Hygrometer
(G) Thermometer

17. A lift starts at floor 0, goes down 7 floors, up 12, down 9, then up 3. Which floor is it on now? [Integers]

(A) Floor 1
(B) Floor 31
(C) Floor 3
(D) Floor -7
(E) Floor -1
(F) Floor 0
(G) Floor -31

18. Express 3/8 as a decimal. [A Peek Beyond the Point]

(A) 0.275
(B) 0.125
(C) 0.375
(D) 3.8
(E) 0.38
(F) 2.667
(G) 0.83

19. A candle of mass 50 g burns, and 35 g of it is used up in 70 minutes. What is the rate at which it burns, in grams per minute? [Physical & Chemical Changes]

(A) 0.35 g/min
(B) 0.5 g/min
(C) 2 g/min
(D) 1.4 g/min
(E) 1 g/min
(F) 5 g/min

20. Rainfall over five days was 12, 0, 8, 20 and 10 mm. What was the average daily rainfall? [Weather, Climate & Adaptations]

- **(A)** 5 mm
- **(B)** 50 mm
- **(C)** 15 mm
- **(D)** 40 mm
- **(E)** 12.5 mm
- **(F)** 10 mm
- **(G)** 8 mm

21. Evaluate (-3) x (-3) x (-3) + (-2) x (-2). [Integers]

- **(A)** 31
- **(B)** -31
- **(C)** -27
- **(D)** 13
- **(E)** -23
- **(F)** -13

22. A rectangle measures 2.4 m by 1.35 m. What is its perimeter? [A Peek Beyond the Point]

- **(A)** 6.5 m
- **(B)** 7.5 m
- **(C)** 8.5 m
- **(D)** 7.05 m
- **(E)** 15 m
- **(F)** 3.24 m

23. Why is the souring of milk into curd a chemical change? [Physical & Chemical Changes]

- **(A)** The milk can easily be turned back into fresh milk
- **(B)** The milk only changes its temperature
- **(C)** No new substance is ever formed
- **(D)** A new substance, curd, is formed and the change cannot be reversed
- **(E)** It is just milk in a different shape
- **(F)** It is purely a change of state
- **(G)** Only the colour changes, nothing else

24. At what time of day is the maximum temperature usually reached? [Weather, Climate & Adaptations]

- **(A)** In the early afternoon, a little after noon
- **(B)** It is the same all day
- **(C)** Late at night
- **(D)** At midnight
- **(E)** Exactly at sunrise
- **(F)** Exactly at noon
- **(G)** Just before dawn

25. At dawn the temperature was -6 C, by noon it rose to 11 C, and by midnight it fell to -9 C. Through how many degrees in total did the temperature move during the day? [Integers]

- **(A)** 3 degrees
- **(B)** 5 degrees
- **(C)** 17 degrees
- **(D)** 37 degrees
- **(E)** 20 degrees
- **(F)** 11 degrees
- **(G)** 26 degrees

26. Find 0.2 of 0.2 of 50. [A Peek Beyond the Point]

- **(A)** 10
- **(B)** 2
- **(C)** 5
- **(D)** 4
- **(E)** 20
- **(F)** 0.2
- **(G)** 2.5

27. Photosynthesis in a leaf is considered a chemical change because: [Physical & Chemical Changes]

- **(A)** The leaf simply gets warmer
- **(B)** Water merely evaporates from the leaf
- **(C)** The leaf only changes its shape
- **(D)** It can be easily reversed by cooling
- **(E)** New substances such as glucose and oxygen are formed
- **(F)** Sunlight only changes the leaf's colour
- **(G)** No new substance is produced

28. Why are tropical rainforests home to such a huge variety of plants and animals? [Weather, Climate & Adaptations]

- **(A)** They are frozen for most of the year
- **(B)** They have no rainfall at all
- **(C)** They are warm and receive plenty of rainfall throughout the year
- **(D)** They are cold and dry all year
- **(E)** They have salty soil and no water
- **(F)** They are dark and lifeless
- **(G)** They get sunlight only once a year

29. A diver descends 18 m, rises 7 m, then descends 11 m more. Where is the diver relative to the surface? [Integers]

- **(A)** 36 m below the surface
- **(B)** 36 m above the surface
- **(C)** 2 m below the surface
- **(D)** 18 m below the surface
- **(E)** 22 m above the surface
- **(F)** 22 m below the surface
- **(G)** 14 m below the surface

30. 73.50 rupees is shared equally among 6 friends. How much does each receive? [A Peek Beyond the Point]

- **(A)** 14.70 rupees
- **(B)** 11.25 rupees
- **(C)** 12.20 rupees
- **(D)** 13.25 rupees
- **(E)** 12.25 rupees
- **(F)** 12.05 rupees

31. Why do we obtain pure copper sulphate crystals by crystallisation rather than by boiling the solution dry? [Physical & Chemical Changes]

- **(A)** Boiling dry is the only way to get crystals
- **(B)** Crystallisation adds new chemicals to the salt
- **(C)** Both methods are exactly the same
- **(D)** Crystallisation needs no water at all
- **(E)** Boiling dry keeps the crystals coloured better
- **(F)** Boiling dry always gives bigger pure crystals
- **(G)** Slow crystallisation gives large pure crystals, while boiling dry can spoil them and trap impurities

32. Humidity readings on four days were 60, 65, 70 and 55. A fifth reading makes the five-day average 64. What was the fifth reading? [Weather, Climate & Adaptations]

- **(A)** 64
- **(B)** 50
- **(C)** 70
- **(D)** 75
- **(E)** 80
- **(F)** 68

33. Evaluate (-144) / 12 / (-3). [Integers]

- **(A)** 4
- **(B)** 40
- **(C)** -12
- **(D)** 48
- **(E)** 12
- **(F)** -48
- **(G)** -4

34. Round 7.2845 to two decimal places. [A Peek Beyond the Point]

- **(A)** 7.29
- **(B)** 7.30
- **(C)** 7.284
- **(D)** 7.2
- **(E)** 7.28
- **(F)** 7.3

35. 8 g of hydrogen reacts completely with 64 g of oxygen to form water. By conservation of mass, what mass of water is formed? [Physical & Chemical Changes]

- **(A)** 64 g
- **(B)** 36 g
- **(C)** 56 g
- **(D)** 72 g
- **(E)** 8 g
- **(F)** 80 g

36. Penguins often huddle tightly together in large groups. How does this help them? [Weather, Climate & Adaptations]

- **(A)** It keeps them apart from danger only
- **(B)** It allows them to dig burrows faster
- **(C)** It helps them catch more sunlight to fly
- **(D)** It helps them lose extra heat in the cold
- **(E)** It increases the surface losing heat
- **(F)** It reduces heat loss and keeps them warm in the cold
- **(G)** It cools them down in the freezing wind

37. Find the sum of all integers from -5 to 6 inclusive. [Integers]

- **(A)** 6
- **(B)** 0
- **(C)** 1
- **(D)** 5
- **(E)** 11
- **(F)** -6

38. Which number is the largest: 0.5, 0.45, 0.405, 0.54, 0.045? [A Peek Beyond the Point]

- **(A)** 0.45
- **(B)** 0.5
- **(C)** 0.405
- **(D)** 0.54
- **(E)** 0.054
- **(F)** 0.045
- **(G)** 0.504

39. Galvanisation protects iron from rusting by coating it with a layer of: [Physical & Chemical Changes]

- **(A)** Gold
- **(B)** Zinc
- **(C)** Copper
- **(D)** Aluminium oxide
- **(E)** Rust
- **(F)** Plastic paint only
- **(G)** Carbon

40. Which best describes the climate of a hot desert? [Weather, Climate & Adaptations]

- **(A)** Snowy all year with little sunshine
- **(B)** Very low rainfall, with hot days and a large difference between day and night temperatures
- **(C)** Cool days and warm nights with constant rain
- **(D)** Mild and rainy throughout the year
- **(E)** Freezing cold and very wet
- **(F)** Heavy rainfall all year and steady temperatures

41. In a quiz each correct answer scores +4 and each wrong answer scores -2. Out of 20 questions a student gets 13 correct and the rest wrong. What is the total score? [Integers]

- **(A)** 24
- **(B)** 40
- **(C)** -14
- **(D)** 38
- **(E)** 66
- **(F)** 52
- **(G)** 26

42. A car uses 0.08 litre of fuel per km. How much fuel does it need for 350 km? [A Peek Beyond the Point]

- **(A)** 28 litres
- **(B)** 0.28 litres
- **(C)** 2.8 litres
- **(D)** 280 litres
- **(E)** 32 litres
- **(F)** 35 litres

43. Rusting of iron needs which two things to be present together? [Physical & Chemical Changes]

- **(A)** Nitrogen and water
- **(B)** Oxygen (air) and water (moisture)
- **(C)** Water and salt only
- **(D)** Only dry air
- **(E)** Heat and light only
- **(F)** Oil and air
- **(G)** Oxygen and carbon dioxide

44. A region's annual rainfall falls from 1200 mm to 900 mm. What is the percentage decrease? [Weather, Climate & Adaptations]

- **(A)** 300%
- **(B)** 20%
- **(C)** 30%
- **(D)** 25%
- **(E)** 33%
- **(F)** 12%

45. Evaluate (-1) x (-2) x (-3) x (-4) x (-5). [Integers]

(A) -15
(B) -120
(C) 60
(D) 0
(E) 120
(F) 15
(G) -60

46. Divide 9.6 / 0.12. [A Peek Beyond the Point]

(A) 0.8
(B) 8
(C) 80
(D) 8.4
(E) 12
(F) 0.08
(G) 800

47. A burning candle shows both kinds of change. The wax melting is a ___ change and the wax burning is a ___ change. [Physical & Chemical Changes]

(A) physical; physical
(B) chemical; chemical
(C) chemical; reversible
(D) neither; neither
(E) physical; chemical
(F) physical; reversible only
(G) chemical; physical

48. Why do many birds migrate to warmer regions during winter? [Weather, Climate & Adaptations]

(A) Because warm regions have no food
(B) To find colder places to nest
(C) To avoid the warm summer heat
(D) Because they cannot fly in summer
(E) To escape the rain only
(F) Because they dislike sunlight
(G) To escape the cold and find enough food

49. The temperatures on three winter nights were -4 C, -7 C and 2 C. What was the average night temperature? [Integers]

(A) -3 C
(B) 3 C
(C) -13 C
(D) -1 C
(E) -9 C
(F) 1 C
(G) 13 C

50. Evaluate 0.125 + 0.0125. [A Peek Beyond the Point]

(A) 0.25
(B) 0.1325
(C) 1.375
(D) 0.2375
(E) 0.1375
(F) 0.15

51. When iron rusts, iron and oxygen combine in a mass ratio of 7 : 3. How much rust forms when 21 g of iron rusts completely? [Physical & Chemical Changes]

(A) 9 g
(B) 21 g
(C) 49 g
(D) 24 g
(E) 28 g
(F) 30 g
(G) 70 g

52. Which features help a camel survive in a hot, dry desert? [Weather, Climate & Adaptations]

(A) Short legs that stay close to hot sand
(B) Gills to breathe underwater
(C) Thick blubber and white fur for snow
(D) Fat stored in its hump, long legs, and the ability to go long without water
(E) A need to drink water every hour
(F) Large ears to keep warm at night

53. If (-15) + box = -23, what number replaces the box? [Integers]

(A) -8
(B) -38
(C) -15
(D) 8
(E) 38
(F) -23

54. A plank 3.6 m long is cut into pieces each 0.45 m long. How many pieces are obtained? [A Peek Beyond the Point]

(A) 80
(B) 8
(C) 9
(D) 1.62
(E) 7
(F) 0.8
(G) 16

55. Which observation is the clearest sign that a chemical change has occurred? [Physical & Chemical Changes]

(A) It is merely moved to a new place
(B) A new substance with new properties is formed
(C) Water on it simply dries up
(D) The material only gets colder
(E) It only changes its size
(F) The material is cut into smaller pieces

56. What does a maximum-minimum thermometer measure? [Weather, Climate & Adaptations]

(A) Only the amount of rainfall
(B) Only the wind speed
(C) The direction of the wind
(D) The brightness of sunlight
(E) The air pressure only
(F) The humidity of the air only
(G) The highest and lowest temperatures reached during a day

57. A submarine at -120 m ascends to -40 m in 4 equal steps. How much does it rise in each step? [Integers]

(A) 20 m down
(B) 40 m up
(C) 20 m up
(D) 30 m up
(E) 80 m up
(F) 160 m up

58. Subtract 5 - 2.376. [A Peek Beyond the Point]

(A) 2.624
(B) 7.376
(C) 2.634
(D) 2.724
(E) 2.376
(F) 2.524
(G) 3.376

59. When carbon dioxide is passed through lime water, the lime water turns milky. Why is this a chemical change? [Physical & Chemical Changes]

(A) The lime water only gets warmer
(B) The water simply evaporates
(C) A new white substance (calcium carbonate) is formed
(D) The gas merely dissolves and can be boiled off
(E) No new substance is produced
(F) It is only a change of state
(G) Only the colour of light changes

60. The temperature rose steadily from 18 C at 6 am to 30 C at noon. What was the average rate of rise per hour? [Weather, Climate & Adaptations]

- **(A)** 3 C per hour
- **(B)** 12 C per hour
- **(C)** 2 C per hour
- **(D)** 1 C per hour
- **(E)** 0.5 C per hour
- **(F)** 6 C per hour
- **(G)** 4 C per hour

61. Evaluate -50 - (-30) + (-15). [Integers]

- **(A)** 5
- **(B)** -65
- **(C)** -35
- **(D)** 65
- **(E)** 35
- **(F)** -95

62. Evaluate 1.2 x 1.2 x 1.2. [A Peek Beyond the Point]

- **(A)** 2.728
- **(B)** 1.628
- **(C)** 3.6
- **(D)** 1.44
- **(E)** 1.728
- **(F)** 1.2
- **(G)** 17.28

63. You have a wooden log. Cutting it into pieces versus burning it - which action forms a new substance? [Physical & Chemical Changes]

- **(A)** Both are physical changes
- **(B)** Only cutting forms new substances
- **(C)** Cutting forms gas, burning does not
- **(D)** Burning the log
- **(E)** Cutting the log
- **(F)** Both form new substances

64. Besides temperature, which factor is most important in deciding the climate of a place? [Weather, Climate & Adaptations]

- **(A)** The brand of houses built there
- **(B)** The amount of moisture, such as humidity and rainfall
- **(C)** The colour of the local soil
- **(D)** The local language spoken
- **(E)** The shape of the roads
- **(F)** The time zone of the region
- **(G)** The number of people living there

65. What is the successor of -1 multiplied by the predecessor of -1? [Integers]

- **(A)** -1
- **(B)** 1
- **(C)** 4
- **(D)** 2
- **(E)** -2
- **(F)** 3
- **(G)** 0

66. A recipe needs 0.35 kg of sugar per cake. How much sugar is needed for 12 cakes? [A Peek Beyond the Point]

- **(A)** 12.35 kg
- **(B)** 42 kg
- **(C)** 4.2 kg
- **(D)** 0.42 kg
- **(E)** 3.5 kg
- **(F)** 4.8 kg

67. A fuel releases 5 kJ of energy per gram when it burns. How much energy is released by burning 0.4 kg of this fuel? [Physical & Chemical Changes]

- **(A)** 5 kJ
- **(B)** 2 kJ
- **(C)** 2000 kJ
- **(D)** 20000 kJ
- **(E)** 500 kJ
- **(F)** 200 kJ

68. Why do many animals living in very cold regions tend to have small ears and short tails? [Weather, Climate & Adaptations]

- **(A)** Larger surface area helps them keep warm
- **(B)** It makes them look bigger to enemies
- **(C)** Smaller body parts reduce the surface area, so they lose less body heat
- **(D)** Small ears help them swim faster
- **(E)** Small parts help them lose heat quickly
- **(F)** Short tails let them fly in the cold

69. A shopkeeper records a loss of 120 rupees on each of 6 days, then a profit of 500 rupees. What is the net result? [Integers]

- **(A)** -1220 rupees
- **(B)** 380 rupees
- **(C)** -120 rupees
- **(D)** 620 rupees
- **(E)** -220 rupees (a loss)
- **(F)** 1220 rupees
- **(G)** 220 rupees (a profit)

70. In the number 24.6573, in which place is the digit 5 and what is its value? [A Peek Beyond the Point]

(A) Thousandths place, worth 0.005
(B) Hundredths place, worth 0.05
(C) Worth 50
(D) Tenths place, worth 0.05
(E) Worth 5
(F) Ten-thousandths place, worth 0.0005
(G) Tenths place, worth 0.5

71. Why is the rusting of an iron bridge a serious problem, and how can it be slowed? [Physical & Chemical Changes]

(A) Rust is harmless, so nothing needs to be done
(B) Rust adds useful weight, so we leave it
(C) Rust makes the iron stronger, so we add more rust
(D) Rusting can only be stopped by cutting the bridge
(E) Rust weakens the iron, so we paint, oil or galvanise it to keep out air and moisture
(F) Rust cools the bridge, so we heat it
(G) Rust is good, so we wet the bridge daily

72. In one week the rainfall totalled 84 mm. If this is a typical week, about how much rain would fall in a year (52 weeks)? [Weather, Climate & Adaptations]

(A) About 436.8 mm
(B) About 1008 mm
(C) About 4368 mm
(D) About 52 mm
(E) About 84 mm
(F) About 4360 mm
(G) About 2184 mm

73. The temperature drops 3 C every 2 hours, starting from 10 C. What is the temperature after 8 hours? [Integers]

(A) -2 C
(B) -12 C
(C) -14 C
(D) 4 C
(E) -22 C
(F) 2 C

74. Evaluate 0.9 - 0.09 - 0.009. [A Peek Beyond the Point]

(A) 0.711
(B) 0.801
(C) 0.81
(D) 0.8
(E) 0.099
(F) 0.819

75. Why is dissolving sugar in water a physical change? [Physical & Chemical Changes]

(A) The sugar changes into salt
(B) The sugar can be recovered by evaporating the water, and no new substance forms
(C) The water turns into sugar
(D) It produces a new gas
(E) A new substance is created in the water
(F) It can never be separated again

76. The plants and animals that live in a region depend most strongly on its: [Weather, Climate & Adaptations]

(A) Local festival dates
(B) Brand of vehicles
(C) Climate
(D) Colour of its flag
(E) Type of currency
(F) Style of clothing shops

77. The product of a number and -7 is 91. What is the number? [Integers]

(A) 84
(B) -13
(C) 13
(D) -7
(E) 98
(F) -98
(G) 7

78. Three rods measure 1.2 m, 0.95 m and 1.45 m. What is their average length? [A Peek Beyond the Point]

(A) 3.6 m
(B) 1.25 m
(C) 1.2 m
(D) 0.9 m
(E) 1.15 m
(F) 1.0 m
(G) 1.3 m

79. Setting of curd, ripening of fruit and digestion of food are all examples of which kind of change? [Physical & Chemical Changes]

(A) Reversible physical changes
(B) No change at all
(C) Changes of state only
(D) Physical changes
(E) Changes of size only
(F) Chemical changes
(G) Changes of shape only

80. A lion's dull tawny (yellow-brown) colour is mainly an adaptation that helps it to: [Weather, Climate & Adaptations]

(A) Blend into a green forest canopy
(B) Stay warm in icy snow
(C) Glow brightly to scare prey
(D) Stay camouflaged in dry grassland while hunting
(E) Shine in the dark to see better
(F) Attract rain to its territory

81. Evaluate (-8) x 5 + 8 x 5. [Integers]

(A) -40
(B) 80
(C) 0
(D) 13
(E) 40
(F) -80

82. Convert 2.05 km into metres. [A Peek Beyond the Point]

(A) 250 m
(B) 2050 m
(C) 205 m
(D) 2.05 m
(E) 2005 m
(F) 20500 m
(G) 2500 m

83. 20 g of iron rusts completely by gaining 8 g of oxygen. What is the mass of rust formed and the percentage increase in mass? [Physical & Chemical Changes]

(A) 12 g of rust, a 40% increase
(B) 40 g of rust, a 40% increase
(C) 20 g of rust, a 40% increase
(D) 28 g of rust, a 28% increase
(E) 28 g of rust, a 40% increase
(F) 28 g of rust, a 60% increase

84. City A averages 25 C with daily temperatures ranging 20-30 C, while City B averages 25 C but ranges 5-45 C. Which city has the more extreme climate and why? [Weather, Climate & Adaptations]

(A) City A, because small ranges are extreme
(B) City B, because it has a much larger temperature range
(C) City A, because its average is higher
(D) City B, because its average is lower
(E) Both are identical because the averages match
(F) City A, because it has the wider range
(G) Neither, because range does not matter

85. A plane flies at 9000 m, descends 2500 m, and a valley floor lies at -300 m (below sea level). How high is the plane above the valley floor? [Integers]

- **(A)** 11500 m
- **(B)** 7100 m
- **(C)** 6200 m
- **(D)** 6800 m
- **(E)** 6500 m
- **(F)** 11800 m
- **(G)** 5900 m

86. Find 0.15 x 240 rupees. [A Peek Beyond the Point]

- **(A)** 36 rupees
- **(B)** 30 rupees
- **(C)** 24 rupees
- **(D)** 3.6 rupees
- **(E)** 16 rupees
- **(F)** 360 rupees
- **(G)** 40 rupees

87. Crystallisation is a ___ change and it is mainly used to ___. [Physical & Chemical Changes]

- **(A)** physical; obtain pure crystals of a solid from its solution
- **(B)** chemical; make rust on iron
- **(C)** chemical; create a brand-new substance
- **(D)** chemical; burn away impurities
- **(E)** physical; turn a solid into a gas forever
- **(F)** physical; mix two solids together
- **(G)** physical; permanently destroy a solid

88. Tropical regions, which are warm throughout the year, lie close to the: [Weather, Climate & Adaptations]

- **(A)** nearest desert only
- **(B)** South Pole
- **(C)** North Pole
- **(D)** Equator
- **(E)** deepest ocean trench
- **(F)** edge of the atmosphere

89. Find the sum (-2) + (-4) + (-6) + ... + (-20). [Integers]

- **(A)** -120
- **(B)** 200
- **(C)** -110
- **(D)** -200
- **(E)** 110
- **(F)** -100
- **(G)** -90

90. Which fraction is equal to 0.6? [A Peek Beyond the Point]

- **(A)** 1/6
- **(B)** 5/3
- **(C)** 3/5
- **(D)** 2/3
- **(E)** 1/60
- **(F)** 6/1000
- **(G)** 3/50

91. When magnesium ribbon is burnt, it gives a bright light and leaves a white powder. This white powder is: [Physical & Chemical Changes]

- **(A)** Pure carbon
- **(B)** Magnesium that has only melted
- **(C)** Magnesium oxide, a new substance
- **(D)** Water vapour
- **(E)** Plain table salt
- **(F)** Unchanged magnesium
- **(G)** Ash from the air

92. Over 30 days, a weather log records 15 sunny, 9 cloudy and 6 rainy days. What fraction of the days were rainy? [Weather, Climate & Adaptations]

- **(A)** 1/3
- **(B)** 6/24
- **(C)** 1/5
- **(D)** 3/10
- **(E)** 1/4
- **(F)** 1/6

93. The product of three integers is -30. Two of them are -5 and 3. What is the third integer? [Integers]

- **(A)** -10
- **(B)** -2
- **(C)** 6
- **(D)** 2
- **(E)** -5
- **(F)** 5
- **(G)** 10

94. Evaluate 0.04 x 0.04. [A Peek Beyond the Point]

- **(A)** 0.08
- **(B)** 16
- **(C)** 0.00016
- **(D)** 0.016
- **(E)** 0.16
- **(F)** 0.0008
- **(G)** 0.0016

95. Why is it usually impossible to easily get back the original substance after a chemical change? [Physical & Chemical Changes]

- **(A)** It just moved to a new container
- **(B)** A new substance with different properties has been formed
- **(C)** The substance only changed size
- **(D)** Only the temperature changed
- **(E)** Nothing new was made, so it is easy to reverse
- **(F)** The original substance only changed shape
- **(G)** It merely changed its state

96. Why do we say weather can change quickly but climate changes only slowly? [Weather, Climate & Adaptations]

- **(A)** Weather is fixed for years but climate changes hourly
- **(B)** Climate is indoors and weather outdoors
- **(C)** Climate is measured every minute, weather every century
- **(D)** Weather is only at night and climate only by day
- **(E)** Weather is the short-term state of the air, while climate is a long-term average that shifts over decades
- **(F)** Both change at exactly the same speed
- **(G)** Neither of them ever changes

97. Which of these has the GREATEST value? [Integers]

- **(A)** (-3) x (-3) = 9
- **(B)** 4 - 9 = -5
- **(C)** (-2) x 5 = -10
- **(D)** (-12) / 3 = -4
- **(E)** 0 x (-7) = 0
- **(F)** (-1) x (-1) x (-1) = -1
- **(G)** (-6) + (-2) = -8

98. You pay with a 100-rupee note for items costing 37.85 rupees and 28.40 rupees. How much change do you get? [A Peek Beyond the Point]

- **(A)** 34.75 rupees
- **(B)** 32.75 rupees
- **(C)** 33.85 rupees
- **(D)** 43.75 rupees
- **(E)** 66.25 rupees
- **(F)** 33.75 rupees
- **(G)** 9.75 rupees

99. A 60 g lump of ice is melted to water and then all the water is boiled away to vapour. Ignoring any loss, what mass of water vapour is produced, and what kind of changes are these? [Physical & Chemical Changes]

- **(A)** 6 g; most mass is destroyed
- **(B)** 60 g of vapour; both are physical changes
- **(C)** 60 g; both are chemical changes
- **(D)** 72 g; vapour weighs more
- **(E)** 30 g; half the mass is lost
- **(F)** 120 g; boiling doubles the mass
- **(G)** 0 g; the water disappears chemically

100. An elephant's very large ears mainly help it to: [Weather, Climate & Adaptations]

(A) Dig tunnels underground

(B) Trap heat to keep warm in snow

(C) Lose extra body heat and stay cool in hot climates

(D) Fly short distances

(E) Shield its eyes from rain

(F) Store food for the winter

(G) Swim across deep rivers